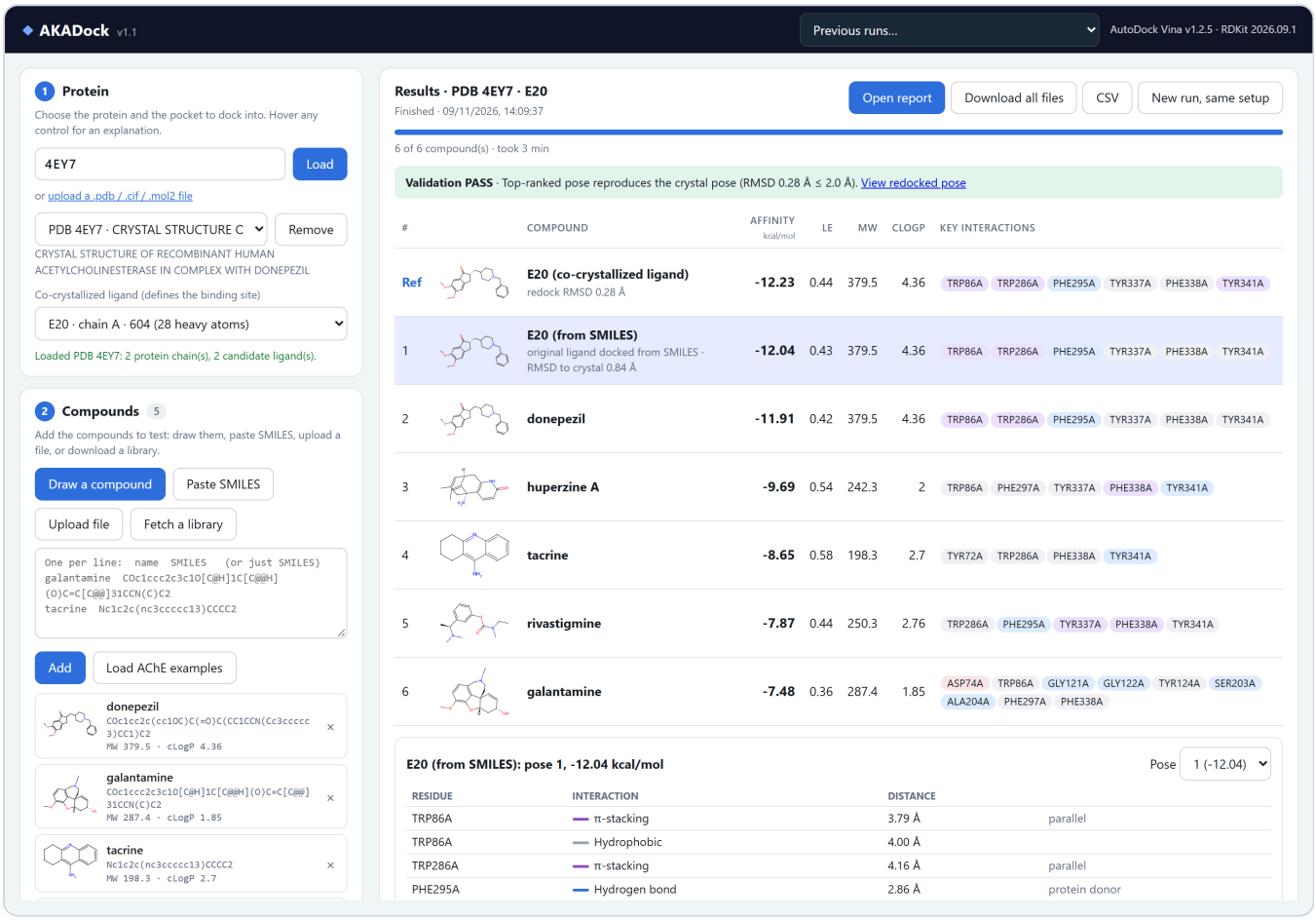
AKADock 1.1
How to dock

A picture-by-picture guide: running your first docking, what every setting does, and what to change when you want something different.



Every screenshot in this guide is a real AKADock 1.1 run: human acetylcholinesterase (PDB 4EY7) with five known inhibitors, docked on an ordinary laptop in two minutes.
AutoDock Vina 1.2.5 · RDKit · PLIP · Open Babel · Ketcher · Mol*

What is in this guide

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2 Step 1 — choose the protein and the pocket

3 Step 2 — add the compounds

4 Step 3 — the settings, one by one

5 While it runs

6 Reading the results

7 The 3D view and the interactions

8 The report and the files

9 “I want this instead” — a recipe table

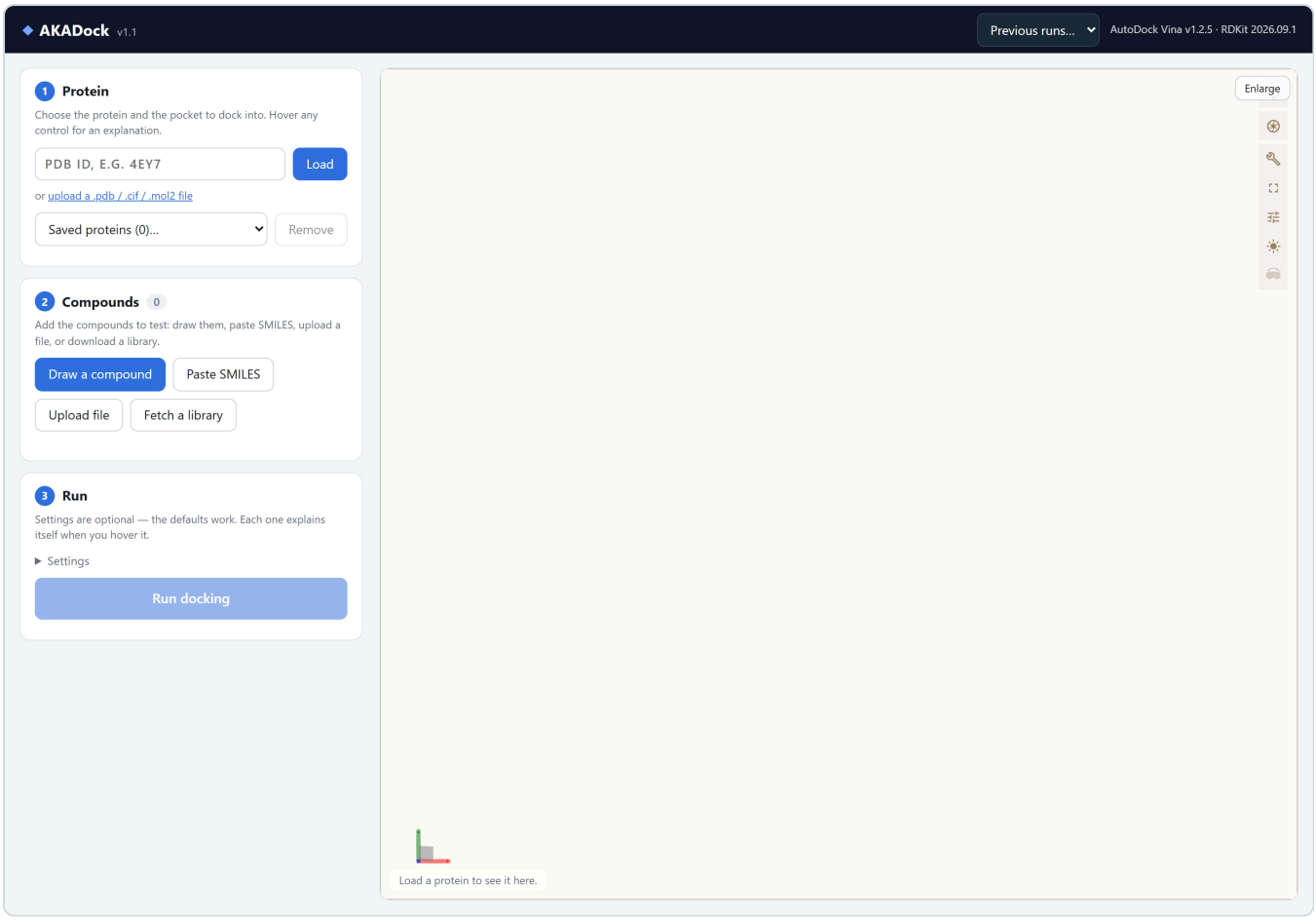
10 When something goes wrong

11 What docking cannot tell you

1 · The idea in one page

Docking asks a narrow question: **if this compound sits in this pocket, what is the best shape it can take, and how well does it fit?** AKADock answers it in three steps you can see on the left of the window — protein, compounds, run — and then checks its own answer before showing it to you.

That check is the part most tools leave out. Before docking your compounds, AKADock takes the ligand that came with the crystal structure, throws it away, and docks it back. If it lands where the crystallographer found it, the setup is sound and the ranking means something. If it does not, you are told so in the same window, in the same run.

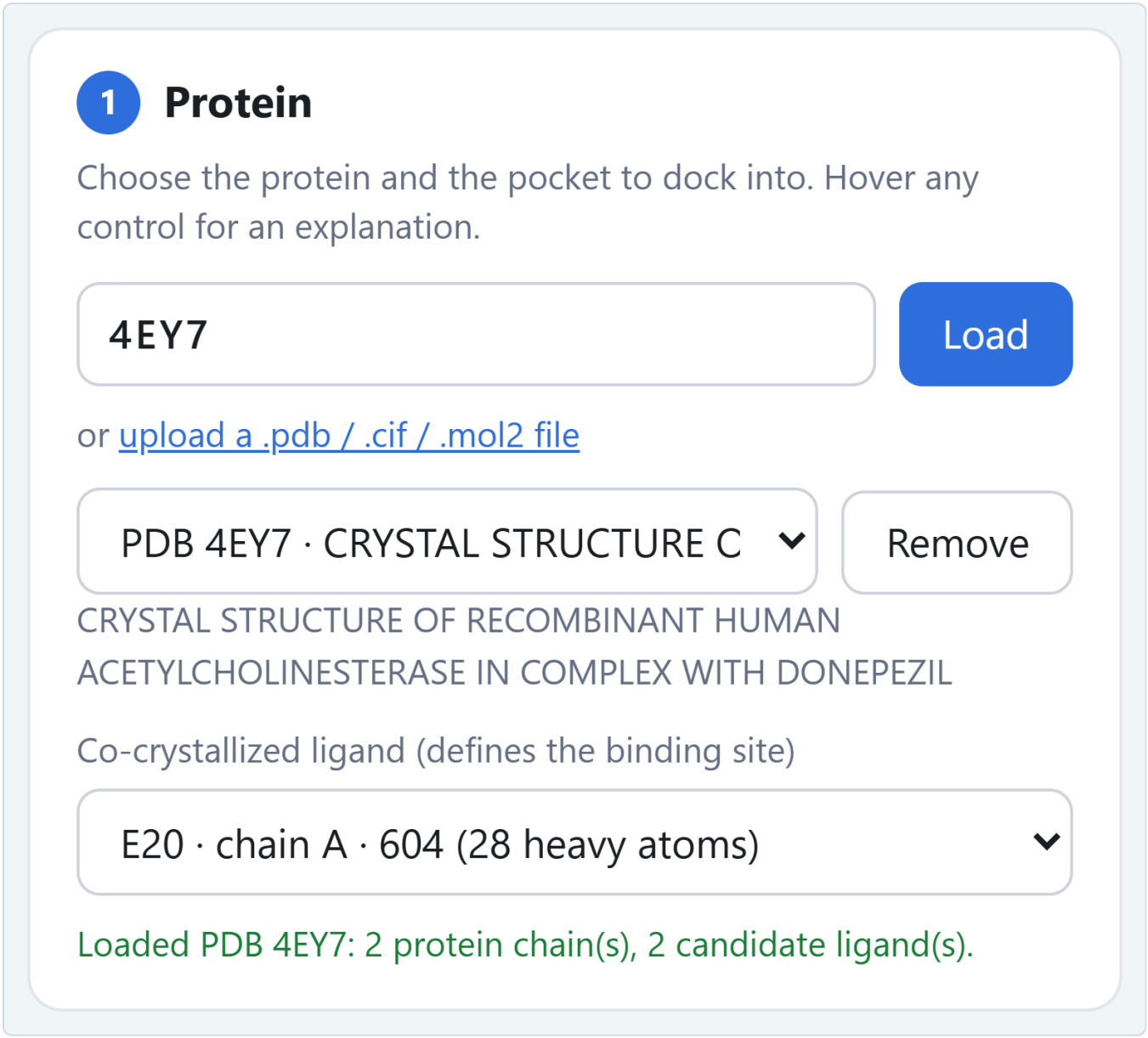


The window when it opens. Three numbered cards on the left, the 3D view on the right. Nothing is hidden in menus: you work down the cards. Hovering any control explains it.

> **Before you start.** Install with `AKADock-Setup.exe` and open AKADock from the Start menu. Windows may warn that the publisher is unknown — click **More info**, then **Run anyway**. Everything runs on your own PC; nothing is uploaded. Only fetching a structure by PDB code or a library from ChEMBL uses the internet.

2 · Step 1 — choose the protein and the pocket

Type a four-character PDB code and press **Load**, or use **upload a .pdb / .cif / .mol2 file** for your own structure. AKADock downloads it, finds the molecules bound inside it, and shows them.



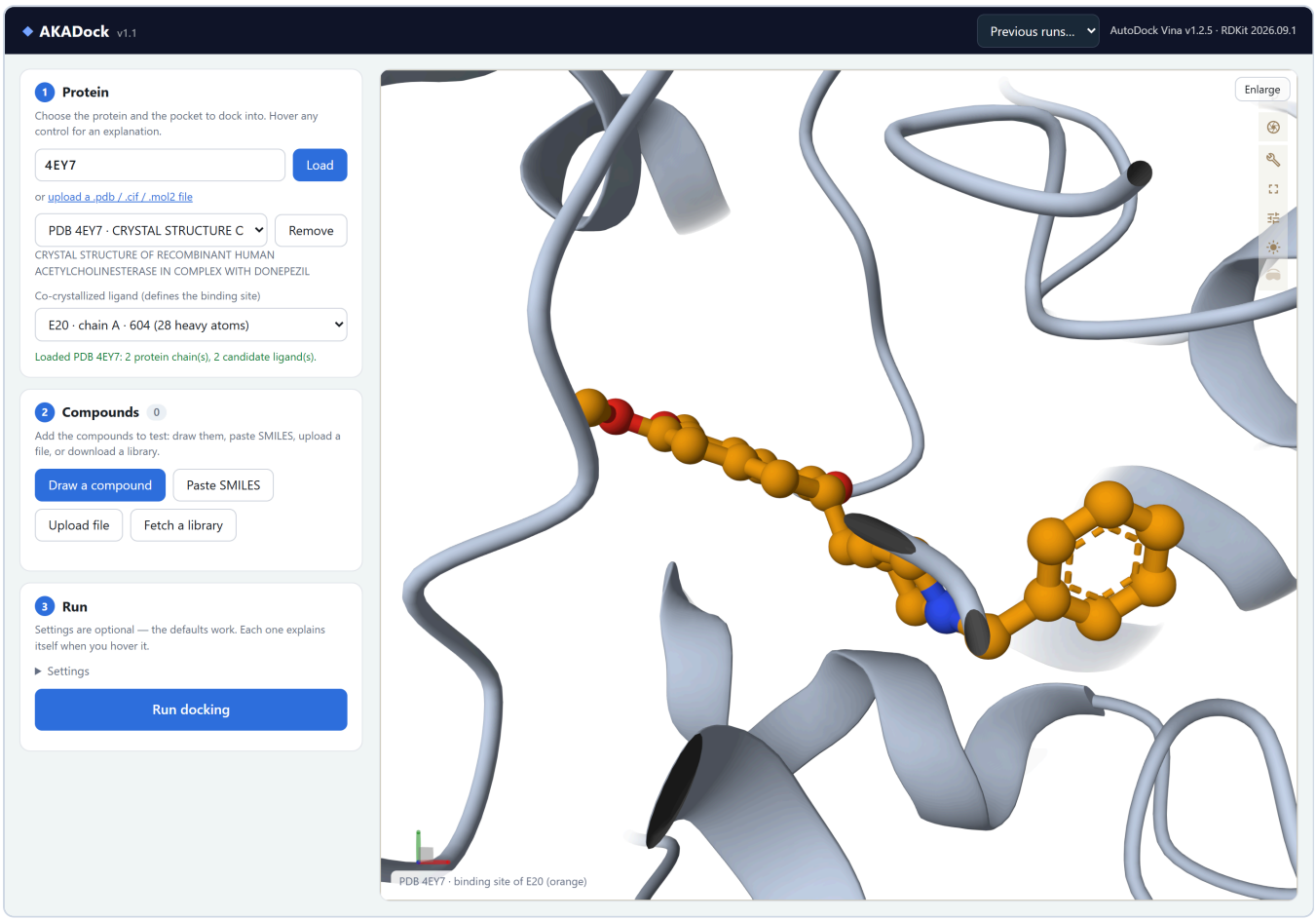
4EY7 loaded. The green line confirms what was found: 2 protein chains and 2 candidate ligands. The dropdown is the important one — the **co-crystallized ligand defines the binding site**. The largest ligand is chosen for you; here that is E20, donepezil, 28 heavy atoms.

The docking box is then built around that ligand, with 10 Å of space around it. Everything else that came with the file — waters, buffer molecules, extra copies — is removed.

If your pocket has no bound ligand

Choose **None** in the dropdown and a text box appears. Type the residues that line the pocket, for example `Trp86 Tyr124 Trp286 Tyr337 Phe338`. One-letter codes (`W86`) and chains (`B:W86`) work too. Press **Show** to see the box before you run.

> **The cost of doing that:** with no crystal ligand there is nothing to redock, so the self-check is skipped and the run cannot tell you whether the setup is trustworthy. Use a structure with a bound ligand whenever you can.



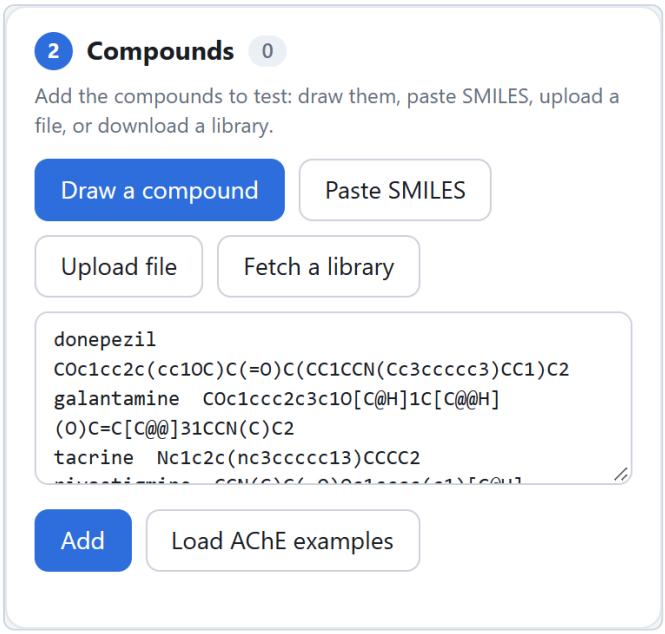
The pocket, before any docking. The bound ligand is orange. Drag to rotate, scroll to zoom, and press **Enlarge** for a bigger view. This is a good moment to check that the ligand you picked really is in the site you care about.

Every protein you load is kept under **Saved proteins**, so the next run with the same target needs no download.

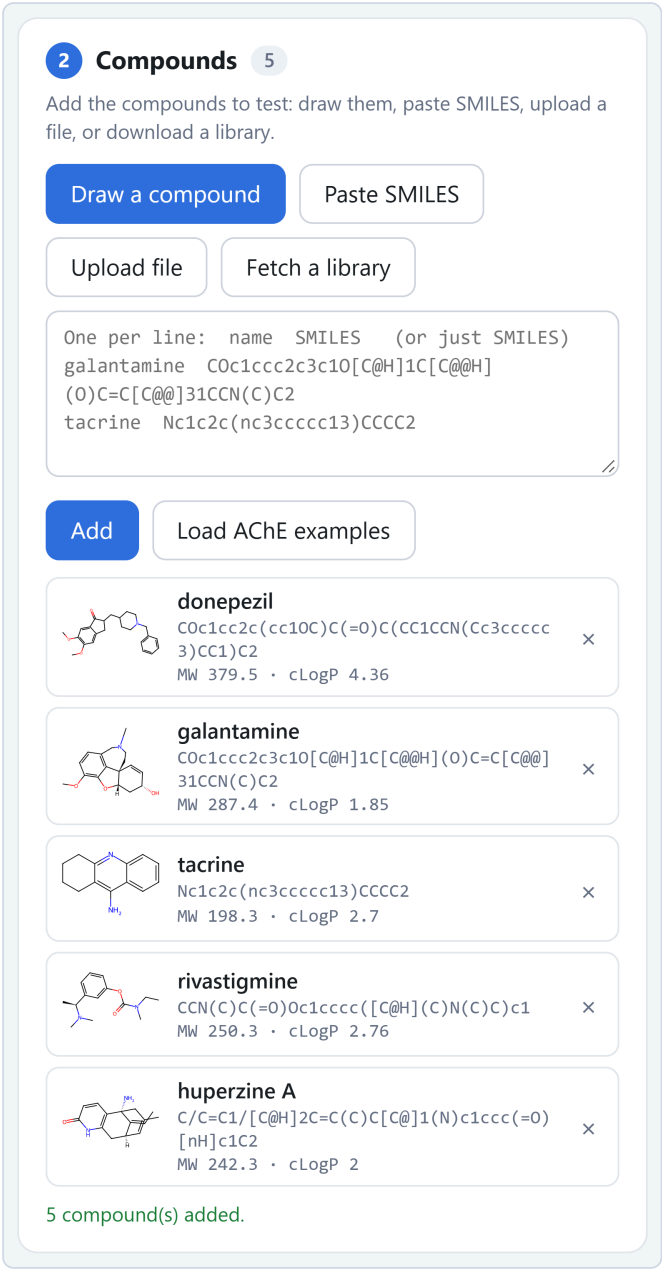
3 · Step 2 — add the compounds

Four ways in, and you can mix them in one run:

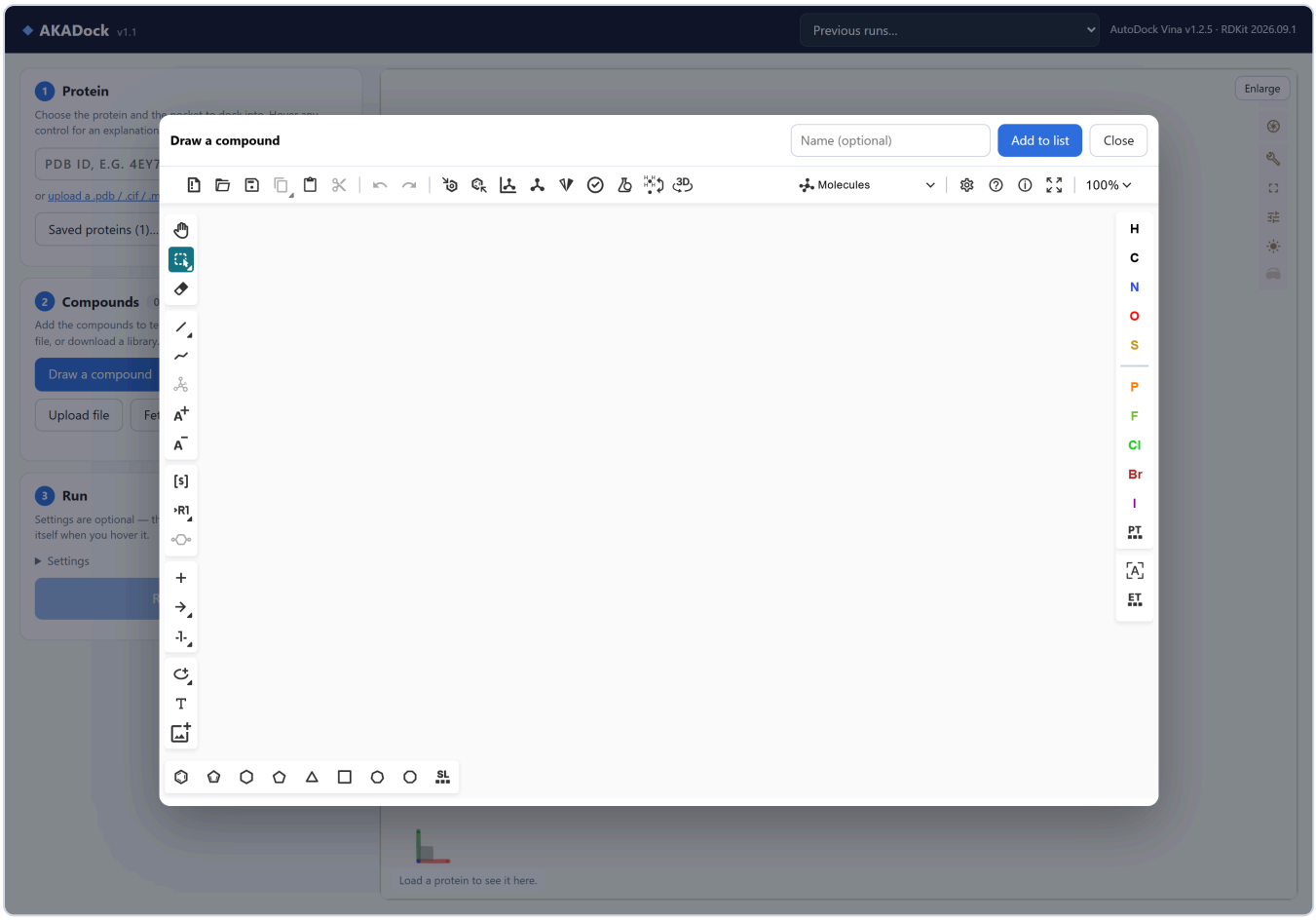
- **Paste SMILES** — one per line, name first then the SMILES, or just the SMILES.
- **Draw a compound** — the Ketcher editor, for when you have a structure on paper.
- **Upload file** — an `.sdf` from PubChem or a vendor, a `.mol2`, or a `.smi` / `.csv` list.
- **Fetch a library** — type a target name (`acetylcholinesterase`) or a ChEMBL id (`CHEMBL220`), pick the species, and the most potent measured compounds are downloaded.



Pasting. Load AChE examples fills this box with five known inhibitors — the quickest way to try the app. Press **Add**.



After adding. Each compound is drawn, named and given its MW and cLogP. Click a name to rename it; the **×** removes one.

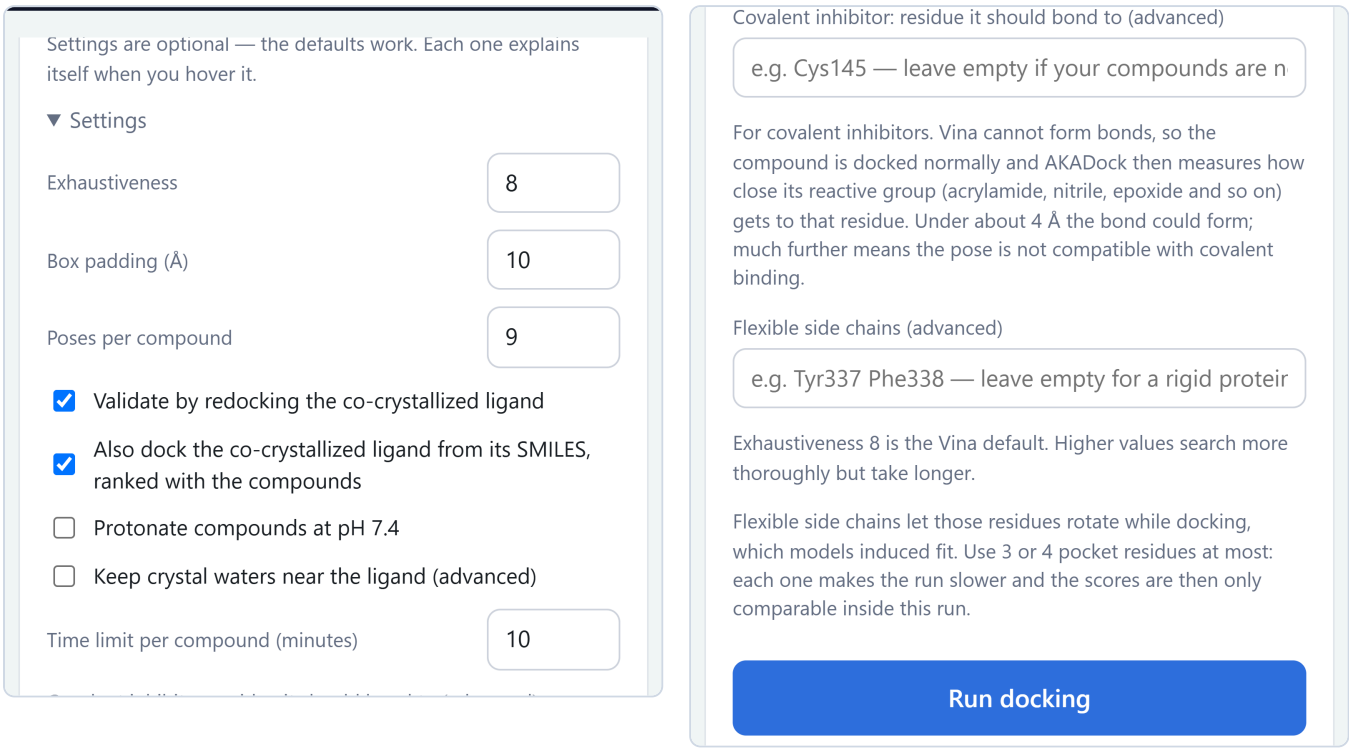


Drawing a compound. Rings and bonds from the toolbars, atoms on the right. Give it a name and press **Add to list**. The structure is converted to SMILES for you.

> **How many at once.** Up to 500 compounds in one run. From 25 compounds AKADock switches to screening mode by itself: everything is still docked, scored and listed, but the slow interaction analysis and the 2D pictures are done afterwards for the 10 best-scoring compounds only.

4 · Step 3 — the settings, one by one

The defaults are the tested ones and most runs need nothing changed. Open **Settings** when you do. Every control explains itself if you hover over it.



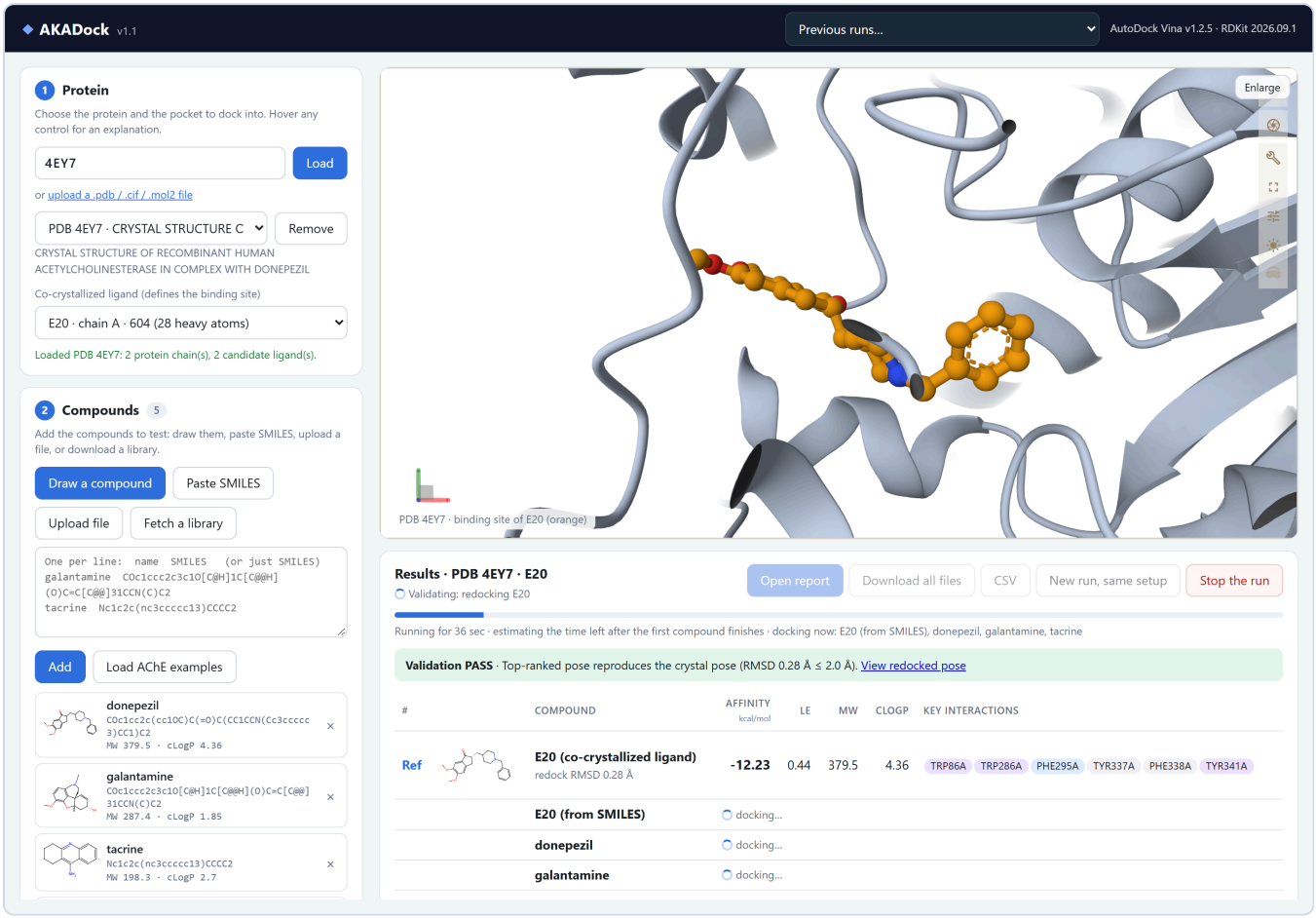
The whole settings panel. Numbers on top, switches in the middle, the two advanced text boxes at the bottom.

Setting	Default	What it does, and when to move it
Exhaustiveness	8	How hard Vina searches. 8 is Vina's own default and is what AKADock's 69 % benchmark was measured at. Raise it to **16–32** for a final, careful run on a few compounds; above about 32 results stop improving. The run time grows roughly in step with the number.
Box padding (Å)	10	How much space around the reference ligand the search box gets. Raise to **12–15** when your compounds are clearly bigger than the ligand that defined the site. A box much larger than the pocket makes the search harder, not better.
Poses per compound	9	How many alternative poses are kept. Lower it to 3–5 for a big screen to save disk space; raise it when you want to study alternatives in the 3D view.
Validate by redocking	on	The self-check described on page 2. **Leave it on.** It costs one extra docking and it is the only thing in the run that tells you whether to believe the ranking.

Also dock the ligand from its SMILES	on	Docks the crystal ligand the same way as your compounds — built in 3D from scratch rather than from crystal coordinates. This is the fair reference to compare your compounds against; the redock alone flatters itself by starting from the right shape.
Protonate at pH 7.4	off	Off on purpose: Vina's scoring has no electrostatics, and an explicitly protonated amine made it flip donepezil in testing. Turn it on only if your protocol requires it.
Keep crystal waters near the ligand	off	Keeps the waters within 3.5 Å of the reference ligand as part of the protein, so a water-mediated contact can be found. Those waters also block space other compounds want: in testing galantamine fell from −7.79 to −5.46 kcal/mol. Use it only for close analogues of the reference ligand.
Time limit per compound	10 min	Gives up on one compound after this long, so a single very flexible molecule cannot block a 500-compound screen. Raise it for large, floppy ligands at high exhaustiveness.
Covalent inhibitor: residue	empty	Type e.g. `Cys145`. The compounds are docked normally, then AKADock finds each warhead (acrylamide, nitrile, epoxide, aldehyde, boronic acid…) and measures how close it gets to that residue. Under about 4 Å the bond could form; much further means the pose is not compatible with covalent binding, however good the score looks.
Flexible side chains	empty	Type 3 or 4 pocket residues (`Tyr337 Phe338`) to let them rotate — induced fit. Eight is the hard limit. Two warnings: scores from a flexible run compare only with each other, and flexible is not automatically better — on 4EY7 the rigid redock is 0.28 Å and the flexible one 0.89 Å, because the crystal side chains are already where they need to be.

5 · While it runs

Press **Run docking**. The run happens in the background: you can keep rotating the 3D view, open an older run, or prepare the next set of compounds.



A run in progress. The validation redock is done first, so the **PASS** verdict appears before your compounds have finished. The bar, the elapsed time and the estimate update as each compound lands; **Stop the run** keeps whatever is already docked and writes the report for it.

Five compounds at the default settings take about two minutes on a four-core laptop. A 500-compound screen is the same work 100 times over — start it and come back. Closing the window stops AKADock, and it asks first if a run is still going.

6 · Reading the results

Validation PASS · Top-ranked pose reproduces the crystal pose (RMSD 0.28 Å ≤ 2.0 Å). View redocked pose

Read this line first.

Verdict	What it means
PASS	The top-ranked redocked pose is within 2.0 Å of the crystal pose. The setup reproduces a known answer, so the ranking is worth reading.
WARN	The crystal pose is among the top three but not ranked first. Vina finds the right answer but does not score it best here — treat small differences between compounds as noise.
FAIL	No top-three pose is within 2.0 Å. Do not rank compounds on this run. Try another structure of the same protein, check you picked the right ligand, raise exhaustiveness, or accept that this pocket is one Vina cannot score (metal sites are the usual reason).

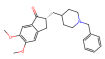
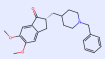
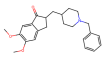
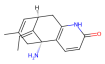
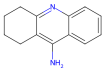
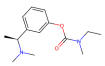
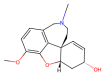
Results · PDB 4EY7 · E20

Finished · 09/11/2026, 14:09:37

Open report | Download all files | CSV | New run, same setup

6 of 6 compound(s) · took 5 min

Validation PASS · Top-ranked pose reproduces the crystal pose (RMSD 0.28 Å ≤ 2.0 Å). View redocked pose

#		COMPOUND	AFFINITY kcal/mol	LE	MW	CLOGP	KEY INTERACTIONS
Ref		**E20 (co-crystallized ligand)** redock RMSD 0.28 Å	**-12.23**	0.44	379.5	4.36	TRP86A TRP286A PHE295A TYR337A PHE338A TYR341A
1		**E20 (from SMILES)** original ligand docked from SMILES · RMSD to crystal 0.84 Å	**-12.04**	0.43	379.5	4.36	TRP86A TRP286A PHE295A TYR337A PHE338A TYR341A
2		**donepezil**	**-11.91**	0.42	379.5	4.36	TRP86A TRP286A PHE295A TYR337A PHE338A TYR341A
3		**huperzine A**	**-9.69**	0.54	242.3	2	TRP86A PHE297A TYR337A PHE338A TYR341A
4		**tacrine**	**-8.65**	0.58	198.3	2.7	TYR72A TRP286A PHE338A TYR341A
5		**rivastigmine**	**-7.87**	0.44	250.3	2.76	TRP286A PHE295A TYR337A PHE338A TYR341A
6		**galantamine**	**-7.48**	0.36	287.4	1.85	ASP74A TRP86A GLY121A GLY122A TYR124A SER203A ALA204A PHE297A PHE338A

E20 (from SMILES): pose 1, -12.04 kcal/mol — Pose 1 (-12.04)

RESIDUE	INTERACTION	DISTANCE	
TRP86A	π-stacking	3.79 Å	parallel
TRP86A	Hydrophobic	4.00 Å	
TRP286A	π-stacking	4.16 Å	parallel
PHE295A	Hydrogen bond	2.86 Å	protein donor
TYR337A	Hydrophobic	3.73 Å	
PHE338A	Hydrophobic	3.53 Å	
PHE338A	Hydrophobic	3.64 Å	
TYR341A	Hydrophobic	3.78 Å	
TYR341A	Hydrophobic	3.57 Å	
TYR341A	Hydrophobic	3.61 Å	
TYR341A	Hydrophobic	3.48 Å	

▶ Log

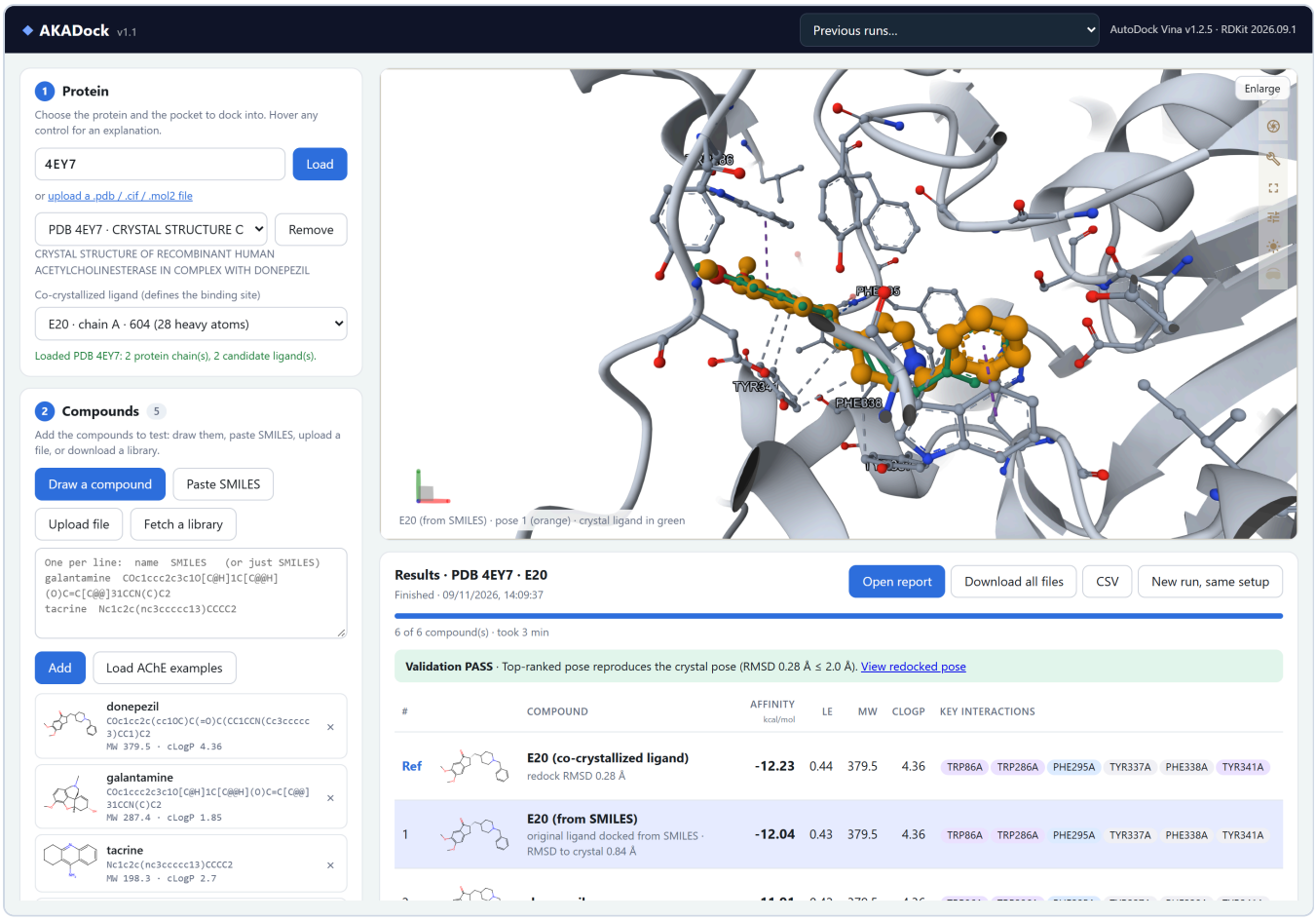
The ranked table. Lower affinity is better. **Ref** is the crystal ligand redocked from its crystal geometry; here the same ligand rebuilt from SMILES lands 0.84 Å from the crystal pose, and donepezil — the same molecule drawn by hand — scores −11.91. Those three lines agreeing is what a healthy run looks like.

Column	How to use it
Affinity	Vina's estimate in kcal/mol, lower is better. Differences under about 1 kcal/mol are inside the method's own error — do not rank on them.
LE	Ligand efficiency: affinity per heavy atom. A small compound with a good LE has more room to grow than a large one with a slightly better score.

MW, cLogP	Molecular weight and lipophilicity, so you can spot a top hit that is simply a big greasy molecule.
Key interactions	The residues the top pose touches, from PLIP. Compare them with the reference ligand's row: a hit that contacts the same residues is more believable than one with a good score somewhere else in the box.

7 · The 3D view and the interactions

Click any row to put that compound in the viewer.



Orange is the docked pose, green is the crystal ligand. Dashed lines are the contacts PLIP found. When the orange sits on the green, the pose is reproducing something real.

E20 (from SMILES): pose 1, -12.04 kcal/mol

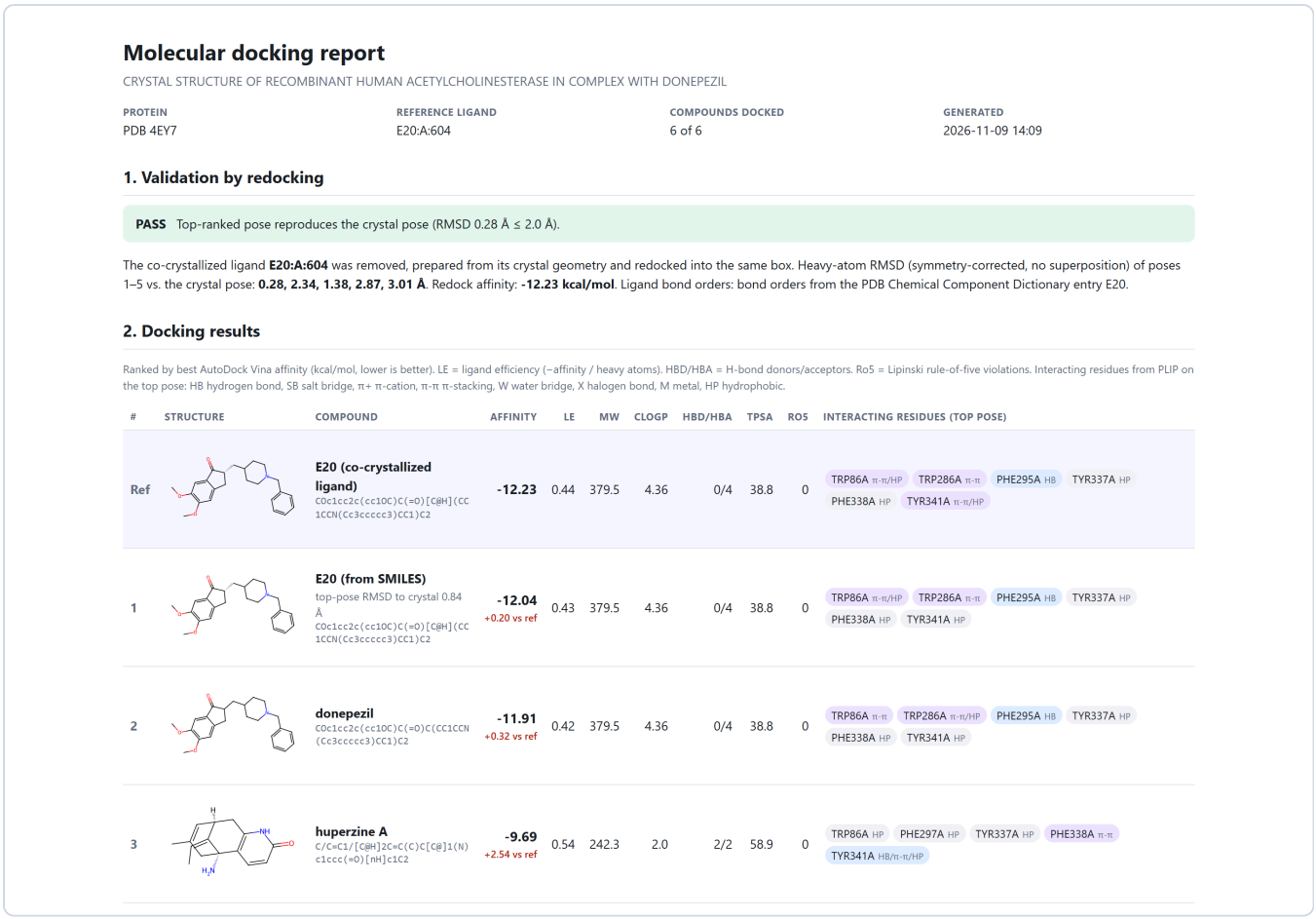
Pose 1 (-12.04)

RESIDUE	INTERACTION	DISTANCE	
TRP86A	π-stacking	3.79 Å	parallel
TRP86A	Hydrophobic	4.00 Å	
TRP286A	π-stacking	4.16 Å	parallel
PHE295A	Hydrogen bond	2.86 Å	protein donor
TYR337A	Hydrophobic	3.73 Å	
PHE338A	Hydrophobic	3.53 Å	
PHE338A	Hydrophobic	3.64 Å	
TYR341A	Hydrophobic	3.78 Å	
TYR341A	Hydrophobic	3.57 Å	
TYR341A	Hydrophobic	3.61 Å	
TYR341A	Hydrophobic	3.48 Å	

Every contact, with its distance. The **Pose** dropdown switches between the alternatives Vina kept — worth a look when the top two scores are close.

8 · The report and the files

Open report writes a single self-contained HTML file: the validation verdict with its numbers, the ranked table with drawn structures, a contact table per compound, and a methods paragraph with every software version. Print it to PDF from your browser if you need one.



Molecular docking report

CRYSTAL STRUCTURE OF RECOMBINANT HUMAN ACETYLCHOLINESTERASE IN COMPLEX WITH DONEPEZIL

PROTEIN	REFERENCE LIGAND	COMPOUNDS DOCKED	GENERATED
PDB 4EY7	E20:A:604	6 of 6	2026-11-09 14:09

1. Validation by redocking

PASS Top-ranked pose reproduces the crystal pose (RMSD 0.28 Å ≤ 2.0 Å).

The co-crystallized ligand **E20:A:604** was removed, prepared from its crystal geometry and redocked into the same box. Heavy-atom RMSD (symmetry-corrected, no superposition) of poses 1–5 vs. the crystal pose: **0.28, 2.34, 1.38, 2.87, 3.01 Å**. Redock affinity: **-12.23 kcal/mol**. Ligand bond orders: bond orders from the PDB Chemical Component Dictionary entry E20.

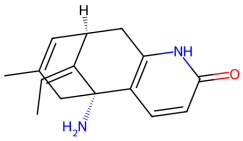
2. Docking results

Ranked by best AutoDock Vina affinity (kcal/mol, lower is better). LE = ligand efficiency (−affinity / heavy atoms). HBD/HBA = H-bond donors/acceptors. Ro5 = Lipinski rule-of-five violations. Interacting residues from PLIP on the top pose: HB hydrogen bond, SB salt bridge, π+ π-cation, π-π π-stacking, W water bridge, X halogen bond, M metal, HP hydrophobic.

#	STRUCTURE	COMPOUND	AFFINITY	LE	MW	CLOGP	HBD/HBA	TPSA	RO5	INTERACTING RESIDUES (TOP POSE)
Ref		**E20 (co-crystallized ligand)** COc1cc2c(cc1OC)C(=O)[C@H](CC1CCN(Cc3ccccc3)CC1)C2	**-12.23**	0.44	379.5	4.36	0/4	38.8	0	TRP86A π-π/HP, TRP286A π-π, PHE295A HB, TYR337A HP, PHE338A HP, TYR341A π-π/HP
1		**E20 (from SMILES)** top-pose RMSD to crystal 0.84 Å COc1cc2c(cc1OC)C(=O)[C@H](CC1CCN(Cc3ccccc3)CC1)C2	**-12.04** +0.20 vs ref	0.43	379.5	4.36	0/4	38.8	0	TRP86A π-π/HP, TRP286A π-π, PHE295A HB, TYR337A HP, PHE338A HP, TYR341A HP
2		**donepezil** COc1cc2c(cc1OC)C(=O)C(CC1CCN(Cc3ccccc3)CC1)C2	**-11.91** +0.32 vs ref	0.42	379.5	4.36	0/4	38.8	0	TRP86A π-π, TRP286A π-π/HP, PHE295A HB, TYR337A HP, PHE338A HP, TYR341A HP
3		**huperzine A** C/C=C1/[C@H]2C=C(C)C[C@]1(N)c1ccc(=O)[nH]c1C2	**-9.69** +2.54 vs ref	0.54	242.3	2.0	2/2	58.9	0	TRP86A HP, PHE297A HP, TYR337A HP, PHE338A π-π, TYR341A HB/π-π/HP

The report. The redock RMSD of each of the first five poses is listed, not just the best one — the honest version of the self-check.

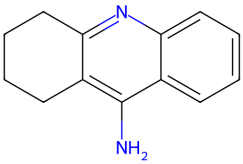
huperzine A · -9.69 kcal/mol



RESIDUE	INTERACTION	DISTANCE (Å)	NOTE
TRP86A	Hydrophobic	3.77	
TRP86A	Hydrophobic	3.98	
TRP86A	Hydrophobic	3.86	
TRP86A	Hydrophobic	3.63	
PHE297A	Hydrophobic	3.67	
TYR337A	Hydrophobic	3.60	
TYR337A	Hydrophobic	3.57	
PHE338A	π-stacking	4.77	T-shaped
TYR341A	Hydrogen bond	4.05	ligand donor
TYR341A	π-stacking	5.04	T-shaped
TYR341A	Hydrophobic	3.97	

All poses (kcal/mol): -9.69, -9.32, -9.27, -8.96, -8.40, -8.08, -7.60, -7.38, -6.83

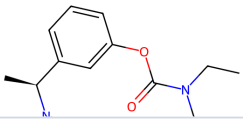
tacrine · -8.65 kcal/mol



RESIDUE	INTERACTION	DISTANCE (Å)	NOTE
TYR72A	Hydrophobic	3.78	
TRP286A	Hydrophobic	3.68	
TRP286A	Hydrophobic	3.78	
TRP286A	Hydrophobic	3.87	
PHE338A	Hydrophobic	3.45	
TYR341A	Hydrogen bond	3.22	ligand donor
TYR341A	π-stacking	4.35	parallel
TYR341A	Hydrophobic	3.64	

All poses (kcal/mol): -8.65, -8.62, -8.55, -8.50, -8.45, -8.17, -8.16, -8.13, -8.05

rivastigmine · -7.87 kcal/mol



RESIDUE	INTERACTION	DISTANCE (Å)	NOTE
TRP286A	Hydrophobic	3.73	
PHE295A	Hydrogen bond	3.67	protein donor
TYR337A	π-stacking	4.87	T-shaped
TYR337A	Hydrophobic	3.80	
PHE338A	π-stacking	4.08	parallel

Further down: the contacts for each compound, and the methods paragraph that tells a reader (or a reviewer) exactly how the numbers were produced.

Download all files gives you a zip with the prepared receptor, every pose as `poses.sdf` and `docked.pdbqt`, complex PDB files ready for PyMOL or Chimera, the CSV and the report. **CSV** alone opens in Excel.

9 · “I want this instead” — a recipe table

What you want	What to change
A more careful answer on a few compounds	Exhaustiveness **16–32**. Expect the run to take two to four times longer.
To screen a few hundred compounds	Leave exhaustiveness at 8, drop **poses per compound** to 3–5, keep the time limit at 10 min. Screening mode switches on by itself from 25 compounds.
My compounds are bigger than the crystal ligand	**Box padding 12–15 Å**. Check in the 3D view that the box still covers the pocket and not half the protein.
The pocket has no ligand in the structure	Ligand dropdown → **None**, then type the pocket residues. Remember the self-check is then skipped.
To model induced fit	**Flexible side chains**: 3 or 4 pocket residues. Compare those scores only with each other.
To test a covalent inhibitor	**Covalent inhibitor: residue** → e.g. `Cys145`. You get a warhead-to-residue distance, not a bond: Vina cannot form bonds (see page 11).
A bridging water is part of the binding	**Keep crystal waters near the ligand** — but only for close analogues of the reference ligand, because those waters block other compounds.
My compound is charged at physiological pH	**Protonate at pH 7.4**. Try both ways and compare; Vina has no electrostatics, so the charged form is not automatically the right input.
To compare my compounds with the known drug	Keep **Also dock the co-crystallized ligand from its SMILES** on, and read your compounds against that line rather than against **Ref**.
To add a few more compounds to a finished run	Add them in step 2 and press **Add to the open run**. Same receptor, same box, same settings, so the scores stay directly comparable.
To dock a different set into the same pocket	**New run, same setup** next to Open report. It copies the protein, site and settings into the form and makes a separate run.
To reopen something from last week	**Previous runs…** at the top. Runs live in the `workspace` folder and are kept when you uninstall.
To run AKADock from Claude	Install the MCP server once: `python mcp\install_mcp.py` (new in 1.1). Claude can then load a protein, dock your compounds and read the results back — and is told to report the validation verdict with every ranking. The docking still happens on this machine.
To work with no internet	Upload a `.pdb` / `.cif` file instead of using a PDB code. Everything else already works offline.

10 · When something goes wrong

What you see	What to do
“Windows protected your PC” when installing	Normal for software without a paid certificate. **More info** → **Run anyway**.
The window opens empty or white	The app window needs the Microsoft Edge WebView2 Runtime, part of Windows 10 and 11. Install it from Microsoft and reopen AKADock.
Validation says FAIL	Not a bug — a result. See page 6. Metal pockets are the most common cause.
A compound is refused	Ligands that contain a metal (heme, metal clusters) cannot be docked by Vina and are refused on purpose. Check the SMILES for a drawing mistake as well.
“Caution” about a metal in the box	Read page 11 before trusting any pose in that site.
One compound runs for ever	It hits the time limit and the run moves on. Very flexible molecules (many rotatable bonds) are simply slow; lower the exhaustiveness or raise the limit.
The structure downloads but has no ligand	Some entries are apo structures. Use the residue list, or find a holo structure of the same protein in the PDB.

> **Where things are.** Runs, saved proteins and the log live in the `workspace` folder next to the program. Uninstalling asks before touching them and keeps them by default.

11 · What docking cannot tell you

AKADock is honest about its limits, and so should any report made with it be.

- **A score is a prediction, not a measurement.** It is not evidence that a compound binds or works. Results have to be confirmed in the laboratory.
- **Differences under ~1 kcal/mol are noise.** Published Vina numbers for the same complex often differ by that much between laboratories.
- **Metal sites are docked without the metal.** Vina has no attractive term for metal coordination, so metal ions are kept for the interaction analysis and the 3D view but are not part of the receptor Vina scores against. Putting them back was tried and measured on 41 metal complexes: redocking got significantly **worse**. A compound that binds a metal — an azole on a heme iron, a hydroxamate on a zinc — can be placed wrongly with a score that looks fine. AKADock warns you when a metal falls inside the box.
- **Covalent binding is not docked.** Vina reads a covalent bond as a clash, so AKADock offers a reactivity check (how close the warhead gets) instead of pretending to form the bond.
- **The protein is rigid** unless you name flexible side chains, and even then the backbone never moves. Missing side-chain atoms in the PDB file are not rebuilt.
- **Protonation, tautomers and ring conformations** are assigned automatically and can be wrong for unusual chemistry.

How well it works, measured

On the Astex Diverse Set — 35 protein–ligand complexes, at AKADock's default settings:

	Top pose ≤ 2 Å	Best of top 3	Median RMSD
Redock (crystal geometry)	**24 / 35 (69 %)**	27 / 35 (77 %)	0.57 Å
Docked from SMILES	**17 / 35 (49 %)**	22 / 35 (63 %)	2.05 Å

Published Vina results on this kind of set are around 58–64 %. The gap between the two rows is the honest cost of starting from a 2D structure — which is what you do with a real lead compound. That is exactly why every run redocks the known ligand and tells you the answer.

AKADock 1.1 · free and open source under the GNU General Public License v2 · the full tables behind these numbers ship with the program in `benchmarks/`, and the reasoning is in `README.md`. Please cite AutoDock Vina, RDKit and PLIP — they do the science.